1. In which phase of the eukaryotic cell cycle does genomic DNA replication occur?
(A) G1 phase
(B) S phase
(C) G2 phase
(D) M phase

2. The quiescent, non-dividing state into which cells withdraw from the cell cycle is termed:
(A) G0 phase
(B) G1 phase
(C) S phase
(D) Apoptotic phase

3. Which regulatory subunit must bind to a Cyclin-Dependent Kinase (CDK) to confer catalytic activity?
(A) Cyclin
(B) Ubiquitin
(C) Calmodulin
(D) Caspase

4. The transition from G1 to S phase across the restriction point is primarily driven by:
(A) Cyclin D-CDK4/6 and Cyclin E-CDK2
(B) Cyclin B-CDK1
(C) Cyclin A-CDK1
(D) Cyclin C-CDK8

5. Which tumor suppressor protein acts as the primary gatekeeper of the G1 restriction point?
(A) Retinoblastoma protein (Rb)
(B) p53
(C) BRCA1
(D) PTEN

6. Hyperphosphorylation of the Rb protein results in the release and activation of:
(A) E2F transcription factor
(B) p53
(C) Cdc25
(D) Myc

7. Which universal cyclin-CDK complex is termed the 'Mitosis Promoting Factor' (MPF)?
(A) Cyclin B - CDK1 (Cdc2)
(B) Cyclin D - CDK4
(C) Cyclin E - CDK2
(D) Cyclin A - CDK2

8. Which protein kinase inhibitory family specifically inhibits CDK4 and CDK6?
(A) INK4 family (p16, p15, p18, p19)
(B) CIP/KIP family (p21, p27)
(C) ATM/ATR
(D) Wee1

9. Which cyclin-dependent kinase inhibitor is a broad-spectrum inhibitor transcriptionally induced by p53?
(A) p21 (CIP1/WAF1)
(B) p16 (INK4a)
(C) p15
(D) p18

10. In which stage of mitosis do sister chromatids condense and the mitotic spindle begins to form?
(A) Prophase
(B) Metaphase
(C) Anaphase
(D) Telophase

11. The breakdown of the nuclear envelope occurs at the transition between:
(A) Prophase and Prometaphase
(B) Metaphase and Anaphase
(C) Anaphase and Telophase
(D) G2 and M

12. Chromosomes align along the equatorial plane of the spindle during:
(A) Prophase
(B) Metaphase
(C) Anaphase
(D) Telophase

13. Synchronous separation of sister chromatids to opposite poles defines which mitotic phase?
(A) Prophase
(B) Prometaphase
(C) Anaphase
(D) Telophase

14. During cytokinesis in animal cells, physical cleavage of the cytoplasm is driven by:
(A) Actin-myosin II contractile ring
(B) Microtubule ring
(C) Cell plate formation
(D) Intermediate filaments

15. In plant cytokinesis, cytoplasmic division occurs from the inside out via assembly of the:

(A) Phragmoplast and cell plate
(B) Contractile ring
(C) Cleavage furrow
(D) Centrosome

16. The multi-protein ring complex that holds duplicated sister chromatids together is:

(A) Cohesin
(B) Condensin
(C) Kinetochore
(D) Centrosome

17. The multi-protein complex that organizes chromosome condensation in prophase is:

(A) Condensin
(B) Cohesin
(C) Kinetochore
(D) Histone H1

18. The specialized proteinaceous structure assembled on the centromere that binds spindle microtubules is:

(A) Kinetochore
(B) Centrosome
(C) Centriole
(D) Telomere

19. The stage of meiosis I where homologous chromosomes begin pairing (synapsis) is:

(A) Leptotene
(B) Zygotene
(C) Pachytene
(D) Diplotene

20. Crossing over and reciprocal genetic recombination between non-sister chromatids occurs in:

(A) Leptotene
(B) Zygotene
(C) Pachytene
(D) Diakinesis

21. X-shaped physical intersections representing sites of crossing over visible in diplotene are:

(A) Chiasmata
(B) Centromeres
(C) Kinetochores
(D) Centrosomes

22. A human somatic cell during G2 phase contains how many chromatids?

(A) 23
(B) 46
(C) 92
(D) 184

23. Meiosis I is characterized as a 'reductional division' because it separates:

(A) Sister chromatids
(B) Homologous chromosomes
(C) Non-homologous chromosomes
(D) Centrioles

24. Meiosis II is characterized as an 'equational division' because it separates:

(A) Sister chromatids
(B) Homologous chromosomes
(C) Bivalents
(D) Tetrads

25. Which enzyme complex is responsible for targeting Cyclin B and Securin for ubiquitin-dependent degradation?

(A) APC/C (Anaphase-Promoting Complex)
(B) SCF complex
(C) Proteasome alone
(D) Caspase-3

26. Separase is an endopeptidase that triggers anaphase onset by cleaving:

(A) Scc1 (Rad21) subunit of Cohesin
(B) Condensin
(C) Tubulin
(D) Cyclin B

27. Before anaphase, Separase is held in an inactive state by binding to its inhibitory chaperone:

(A) Securin
(B) Cohesin
(C) Mad2
(D) BubR1

28. Failure of homologous chromosomes or sister chromatids to separate properly during cell division is:

(A) Non-disjunction
(B) Crossing over
(C) Translocation
(D) Inversion

29. Down syndrome (Trisomy 21) is most commonly caused by meiotic non-disjunction occurring during:

(A) Maternal Meiosis I
(B) Paternal Meiosis II
(C) Mitosis in zygote
(D) S phase

30. Klinefelter syndrome is characterized by which sex chromosome karyotype?

(A) 47, XXY
(B) 45, XO
(C) 47, XYY
(D) 47, XXX

31. Turner syndrome is characterized by which monosomic karyotype?

(A) 45, XO
(B) 47, XXY
(C) 46, XX
(D) 47, XXX

32. Which kinase phosphorylates and inhibits CDK1 to prevent premature mitotic entry during G2?

(A) Wee1 kinase
(B) CAK
(C) Cdc25
(D) Plk1

33. Which dual-specificity phosphatase removes inhibitory phosphates from Thr-14 and Tyr-15 of CDK1?

(A) Cdc25 phosphatase
(B) Wee1
(C) PP2A
(D) Calcineurin

34. CDK-Activating Kinase (CAK) activates CDKs by phosphorylating a conserved threonine in the:

(A) T-loop (Thr-160/161)
(B) ATP-binding pocket
(C) C-terminal tail
(D) N-terminal lobe

35. The tumor suppressor p53 is rapidly degraded in normal unperturbed cells through ubiquitination by:

(A) MDM2 (E3 ubiquitin ligase)
(B) APC/C
(C) SCF
(D) Fbxw7

36. DNA double-strand breaks primarily activate which sensor kinase to initiate DNA damage checkpoint arrest?

(A) ATM kinase
(B) ATR kinase
(C) PKA
(D) MAP kinase

37. Single-stranded DNA coated with RPA at stalled replication forks primarily activates:

(A) ATR kinase
(B) ATM kinase
(C) CDK1
(D) Wee1

38. The Spindle Assembly Checkpoint (SAC) monitors which critical mitotic event before allowing anaphase?

(A) Kinetochore attachment to spindle microtubules and bipolar tension
(B) DNA replication completion
(C) Centrosome duplication
(D) Nuclear envelope reassembly

39. Which soluble protein complex is generated by unattached kinetochores to inhibit APC/C-Cdc20?

(A) Mitotic Checkpoint Complex (MCC: Mad2, BubR1, Bub3, Cdc20)
(B) SCF complex
(C) Cohesin
(D) Condensin

40. The ladder-like proteinaceous structure that zips homologous chromosomes together in pachytene is:

(A) Synaptonemal complex
(B) Cohesin axis
(C) Kinetochore
(D) Spindle matrix

41. The G1 Restriction Point (R-point) in mammalian cells is defined as the point beyond which:

(A) Cell cycle progression becomes independent of extracellular mitogens
(B) DNA replication starts
(C) Chromosomes condense
(D) Apoptosis is irreversible

42. Mitogenic signaling (e.g., EGF via Ras-MAPK) drives progression through early G1 primarily by upregulating:

(A) Cyclin D1
(B) Cyclin B1
(C) p27
(D) Wee1

43. In early G1, hypo-phosphorylated Rb binds E2F and actively represses transcription by recruiting:

(A) Histone Deacetylases (HDACs)
(B) Histone Acetyltransferases (HATs)
(C) DNA Polymerase alpha
(D) RNA Polymerase II

44. Once E2F is released from Rb, it transactivates genes encoding DNA synthesis machinery and:

(A) Cyclin E, Cyclin A, and E2F itself (positive feedback)
(B) p16
(C) Wee1
(D) MDM2 exclusively

45. The transition from G1 to S requires activation of the SCF ubiquitin ligase complex, which degrades:

(A) p27 (Kip1) and Cyclin E
(B) Cyclin B
(C) Securin
(D) Geminin

46. Phosphorylation of p27 on Thr-187 creates a phosphodegron recognized by which F-box protein in the SCF complex?

(A) Skp2
(B) Fbw7
(C) beta-TrCP
(D) Cdc20

47. Origin of Replication licensing occurs exclusively during G1 phase via assembly of the pre-RC consisting of:

(A) ORC, Cdc6, Cdt1, and MCM2-7 helicase
(B) DNA polymerase delta and PCNA
(C) Topoisomerase II
(D) Primase and RPA

48. Re-replication within a single cell cycle is prevented because high S/G2/M CDK activity causes:

(A) Phosphorylation and degradation/inactivation of Cdc6 and Cdt1
(B) Activation of ORC
(C) Dephosphorylation of MCM
(D) Activation of Cdt1

49. Geminin prevents re-licensing of replication origins from S phase through early mitosis by binding to and inhibiting:

(A) Cdt1
(B) Cdc6
(C) ORC1
(D) MCM2

50. At anaphase, Geminin is destroyed by:

(A) APC/C-Cdc20
(B) SCF-Skp2
(C) Proteasome without ubiquitin
(D) Caspase-3

51. The positive feedback loop driving explosive activation of MPF (Cyclin B-CDK1) at G2/M involves:

(A) CDK1 activating Cdc25 and phosphorylating/inhibiting Wee1
(B) CDK1 activating Wee1
(C) CDK1 inhibiting Cdc25
(D) Cyclin D degradation

52. Greatwall kinase (GWL) promotes mitotic entry by phosphorylating Ensa/ARPP19, which inhibits:

(A) PP2A-B55 phosphatase
(B) Cdc25
(C) CDK1
(D) Wee1

53. Phosphorylation of nuclear lamins (Lamin A/B/C) by active CDK1 at the onset of mitosis causes:

(A) Depolymerization of the nuclear lamina and nuclear envelope breakdown
(B) Chromatin condensation
(C) Spindle elongation
(D) Centrosome separation

54. Condensin is activated to compact mitotic chromosomes via phosphorylation by:

(A) Cyclin B - CDK1 and Aurora B
(B) Wee1
(C) E2F
(D) ATM kinase

55. Aurora B kinase is the enzymatic core of the Chromosomal Passenger Complex (CPC), functioning to:

(A) Destabilize erroneous kinetochore-microtubule attachments lacking tension
(B) Promote cytokinesis alone
(C) Phosphorylate tubulin dimers
(D) Synthesize ATP

56. When sister kinetochores achieve correct amphitelic (bipolar) attachment, mechanical tension pulls them apart, which:

(A) Separates Ndc80 from Aurora B, stabilizing the microtubule attachment
(B) Activates Aurora B
(C) Recruits Mad2
(D) Inactivates CDK1

57. The conformational activation of Mad2 at unattached kinetochores involves conversion of:

(A) Open (O-Mad2) to closed (C-Mad2)
(B) Closed to open
(C) Monomer to trimer
(D) Phosphorylated to unphosphorylated

58. The Mitotic Checkpoint Complex (MCC) directly blocks anaphase onset by binding to and inhibiting:

(A) Cdc20, preventing it from activating APC/C
(B) Cdh1
(C) Separase
(D) Cohesin

59. During prophase, the majority of cohesin along chromosome arms is removed via the 'prophase pathway' mediated by:

(A) Wapl and Pds5, triggered by Plk1 and Aurora B phosphorylation
(B) Separase cleavage
(C) Caspase-3
(D) APC/C

60. Centromeric cohesin is protected from premature removal during prophase by:

(A) Shugoshin (Sgo1) recruiting PP2A phosphatase
(B) Wapl alone
(C) Separase
(D) Cdc20

61. In anaphase A, sister chromatid movement toward spindle poles is driven primarily by:

(A) Depolymerization of kinetochore microtubules at plus and minus ends
(B) Spindle pole separation
(C) Actin contraction
(D) Cytokinesis

62. In anaphase B, spindle pole separation is driven primarily by:

(A) Kinesin-5 sliding overlapping interpolar microtubules and cortical dynein pulling astral microtubules
(B) Kinetochore shortening
(C) Cohesin cleavage
(D) Nuclear reassembly

63. During exit from mitosis, the switch from APC/C-Cdc20 to APC/C-Cdh1 is triggered by:

(A) Dephosphorylation of Cdh1 by Cdc14 phosphatase as CDK1 activity falls
(B) Phosphorylation of Cdh1 by CDK1
(C) Cdc20 synthesis
(D) Securin re-accumulation

64. The active GTPase that orchestrates assembly of the contractile actomyosin ring at the cleavage furrow is:

(A) RhoA
(B) Rac1
(C) Cdc42
(D) Ran

65. Centralspindlin (MKLP1 - MgcRacGAP complex) recruits the RhoGEF ECT2 to the spindle midzone to activate:

(A) RhoA at the equatorial cortex
(B) Rac1 at the poles
(C) Cdc42
(D) Rab11

66. Active RhoA at the equatorial cortex stimulates cleavage furrow ingression by activating:

(A) Formin (Dia) to nucleate actin filaments and ROCK to activate Myosin II
(B) Arp2/3 complex
(C) Kinesin-1
(D) Caspases

67. The final step of cytokinesis, physical abscission of the intercellular midbody bridge, is executed by:

(A) The ESCRT-III machinery
(B) Dynamin
(C) SNARE complexes alone
(D) Actin ring contraction alone

68. In DNA damage signaling, ATM phosphorylates the variant histone H2AX on Ser-139, forming:

(A) gamma-H2AX, which serves as a recruitment platform for DNA repair complexes
(B) Ubiquitinated H2AX
(C) Methylated H2AX
(D) Cleaved H2AX

69. ATM and ATR arrest the cell cycle by phosphorylating Chk2 and Chk1 kinases, which in turn phosphorylate and inactivate:

(A) Cdc25 phosphatases, targeting them for degradation or 14-3-3 sequestration
(B) Wee1 kinase
(C) p53
(D) p16

70. Phosphorylation of p53 on Ser-15 and Ser-20 by ATM/Chk2 disrupts its interaction with:

(A) MDM2, stabilizing p53 and preventing its proteasomal degradation
(B) p21
(C) DNA
(D) Bax

71. When severe DNA damage cannot be repaired, p53 transactivates pro-apoptotic genes including:

(A) Puma, Noxa, and Bax
(B) Bcl-2 and Bcl-xL
(C) Cyclin D1
(D) MDM2 exclusively

72. The synaptonemal complex transverse filaments in mammals are composed predominantly of:

(A) SYCP1
(B) SYCP2
(C) SYCP3
(D) Rad51

73. Meiotic double-strand breaks that initiate crossing over are deliberately catalyzed by the topoisomerase-like transesterase:

(A) Spo11
(B) Topoisomerase II
(C) Cas9
(D) RAG1/RAG2

74. In Meiosis I, the protection of centromeric cohesin (Rec8) from separase cleavage is executed by:

(A) Shugoshin (Sgo2) recruiting PP2A to the centromere
(B) Wapl
(C) Mad2
(D) Bub1

75. Degradation of centromeric Rec8 cohesin in Meiosis II allows separation of:

(A) Sister chromatids
(B) Homologous chromosomes
(C) Bivalents
(D) Polar bodies

76. The catalytic cleft of CDK2 requires two simultaneous structural rearrangements for activation: T-loop displacement and:

(A) Rotation of the PSTAIRE alpha-helix to align catalytic Glu-51 with Lys-33
(B) Cleavage of the N-terminus
(C) Dimerization of CDK2
(D) Binding of zinc

77. In unphosphorylated, monomeric CDK2, the T-loop physically blocks the active site cleft by projecting across the:

(A) Substrate-binding pocket
(B) ATP-binding pocket
(C) Cyclin interface
(D) C-terminal lobe

78. Phosphorylation of Thr-160 on CDK2 by CAK stabilizes the active conformation by forming an electrostatic triad with:

(A) Arg-50, Arg-126, and Arg-150
(B) Asp-145 and Glu-51
(C) Lys-33 and Asp-166
(D) Two magnesium ions

79. The structural basis of INK4 (p16) inhibition of CDK4/6 is that INK4 binding to the opposite face of the catalytic cleft:

(A) Distorts the ATP-binding pocket and misaligns the cyclin D binding interface
(B) Phosphorylates the T-loop
(C) Cleaves Cyclin D
(D) Occludes the substrate site directly

80. CIP/KIP inhibitors (p21, p27) insert an extended N-terminal inhibitory domain directly into the:

(A) ATP-binding pocket of CDK2, mimicking the adenine ring and inserting an inhibitory peptide
(B) Cyclin groove alone
(C) T-loop exclusively
(D) Nuclear export signal

81. The Ndc80 complex is a 57 nm rod-like heterotetramer (Ndc80-Nuf2-Spc24-Spc25) whose N-terminal globular domain binds microtubules via:

(A) A basic, positively charged 80-amino-acid tail and calponin homology (CH) domains
(B) An SH2 domain
(C) A zinc finger
(D) A kinase domain

82. Aurora B kinase destabilizes low-tension kinetochore-microtubule attachments by phosphorylating how many serine/threonine sites on the Ndc80 N-terminal tail?

(A) Up to 9 conserved basic-patch phosphorylation sites
(B) Exactly 1 site
(C) 2 sites
(D) 15 sites

83. Phosphorylation of the Ndc80 tail by Aurora B introduces negative charges that:

(A) Electrostatically repel the negatively charged E-hook of tubulin, dropping binding affinity
(B) Increase tubulin binding
(C) Crosslink Ndc80
(D) Sever microtubules

84. In the Mad2 conformational template model, ligand-bound Closed Mad2 (C-Mad2) at the kinetochore recruits soluble Open Mad2 (O-Mad2) and:

(A) Catalyzes its conversion into C-Mad2 bound to Cdc20
(B) Phosphorylates O-Mad2
(C) Degrades O-Mad2
(D) Cleaves Cdc20

85. The Mitotic Checkpoint Complex (MCC) binds the Anaphase-Promoting Complex (APC/C) as a pseudo-substrate inhibitor by occupying:

(A) Both the substrate-binding co-receptor site on Cdc20 and the core catalytic groove of APC/C
(B) The E2 binding pocket alone
(C) The cullin ring alone
(D) The RNA binding cleft

86. p53 functions as a sequence-specific transcription factor that binds DNA as a:

(A) Tetramer (dimer of dimers) recognizing two decameric consensus half-sites
(B) Monomer
(C) Heterodimer with Myc
(D) Hexamer

87. The oncogenic E3 ubiquitin ligase MDM2 binds to the extreme N-terminal transactivation domain of p53 via a deep hydrophobic pocket recognizing:

(A) Phe-19, Trp-23, and Leu-26 of p53
(B) Ser-15 and Ser-20
(C) Lys-382
(D) The tetramerization domain

88. Nutlin-3 is a potent small-molecule anticancer drug that specifically restores p53 function by:

(A) Occupying the hydrophobic pocket of MDM2, competitively displacing p53
(B) Inactivating p53
(C) Inhibiting ATM kinase
(D) Degrading p21

89. The 'Restriction Point' model of Pardee establishes that normal cells become committed to S-phase entry when:

(A) Rb phosphorylation exceeds 80% and E2F transcriptional activity reaches a bistable bistable threshold
(B) p53 is degraded
(C) CDK1 is synthesized
(D) Centrosomes duplicate

90. In the G2 DNA damage checkpoint, Claspin serves as an essential mediator that couples ATR to the phosphorylation and activation of:

(A) Chk1 kinase on Ser-317 and Ser-345
(B) Chk2 kinase
(C) p53 directly
(D) Wee1 directly

91. The 14-3-3 sigma protein enforces G2/M arrest in response to DNA damage by:

(A) Binding phosphorylated Ser-216 of Cdc25C and sequestering it in the cytoplasm
(B) Inactivating Cyclin B
(C) Degrading CDK1
(D) Cleaving tubulin

92. During cytokinesis, the abscission checkpoint (NoCut checkpoint) delays final membrane cleavage if lagging chromatin is detected in the bridge via activation of:

(A) Aurora B kinase maintaining active ESCRT-III phosphorylation
(B) Plk1
(C) RhoA
(D) Caspase-3

93. Cohesin rings topologically entrap sister chromatid DNA within a tripartite ring formed by:

(A) Smc1, Smc3, and the kleisin subunit Scc1 (Rad21)
(B) Smc2 and Smc4
(C) Condensin I and II
(D) Histone octamers

94. During anaphase onset, Separase cleaves Scc1 at how many specific consensus peptide sites?

(A) Two conserved arginine sites (Arg-172 and Arg-450 in humans)
(B) One site
(C) Five sites
(D) Randomly

95. The kinetochore-localized motor protein that drives poleward chromosome motion while coupled to depolymerizing plus-ends is:

(A) CENP-E (Kinesin-7) and Dynein
(B) Kinesin-1
(C) Myosin II
(D) Kinesin-5

96. The Dam1/DASH complex in budding yeast forms a complete 16-membered ring around the microtubule that:

(A) Acts as a sliding collar harnessing the mechanical energy of curling protofilaments to pull chromosomes
(B) Sever microtubules
(C) Caps the minus-end
(D) Hydrolyzes ATP

97. In human cells, the functional equivalent of the Dam1 ring harnessing microtubule depolymerization energy is the:

(A) Ndc80 complex working cooperatively with the Ska1 complex
(B) Centrosome
(C) Cohesin ring
(D) Condensin ring

98. Meiotic homologous recombination is initiated by Spo11, which cleaves both DNA strands via a covalent intermediate linking DNA to:

(A) A catalytic Tyrosine residue (Tyr-98)
(B) A Serine residue
(C) A Lysine residue
(D) A Histidine residue

99. Resolution of double Holliday junctions (dHJs) in meiotic pachytene to generate obligate crossover products is catalyzed by:

(A) The MutL-gamma (Mlh1-Mlh3) resolvase endonuclease complex
(B) RecA alone
(C) DNA ligase I alone
(D) Topoisomerase I

100. Mosaic Variegated Aneuploidy (MVA) is an autosomal recessive human syndrome characterized by widespread aneuploidy caused by biallelic mutations in:

(A) BUB1B (encoding BubR1 checkpoint kinase)
(B) TP53
(C) RB1
(D) CDK1

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the biochemical architecture of Cyclin-Dependent Kinases (CDKs) and detail the dual structural mechanism of CDK activation by Cyclin binding and CAK phosphorylation.

Q2. Describe the Retinoblastoma (Rb) - E2F pathway governing the G1/S restriction point and explain how bistable switching ensures irreversible S-phase commitment.

Q3. Contrast the two families of Cyclin-Dependent Kinase Inhibitors (INK4 and CIP/KIP) regarding genetic loci, structural mechanisms, and specificities.

Q4. Detail the molecular mechanism of Pre-Replication Complex (pre-RC) licensing during G1 and explain how re-replication is prevented.

Q5. Explain the bistable biochemical circuitry of Mitosis Promoting Factor (MPF / Cyclin B-CDK1) activation at the G2/M transition.

Q6. Describe the structural organization and function of the Kinetochore, detailing the inner KMN network and the Ndc80 complex.

Q7. Explain the Error Correction mechanism of Kinetochore-Microtubule Attachments mediated by Aurora B Kinase and Tension Sensing.

Q8. Detail the molecular mechanism of the Spindle Assembly Checkpoint (SAC), explaining MCC generation and APC/C-Cdc20 inhibition.

Q9. Explain the molecular cascade triggering Anaphase Onset, detailing APC/C activation, Securin degradation, and Cohesin cleavage by Separase.

Q10. Describe the stages of Meiosis I and II, highlighting the molecular architecture of the Synaptonemal Complex and Chiasma resolution.

Q11. Explain the molecular mechanics of Cytokinesis in animal cells, detailing Centralspindlin, ECT2, RhoA, the Contractile Ring, and ESCRT-III Abscission.

Q12. Detail the DNA Damage Checkpoint signaling cascade mediated by ATM/ATR, Chk1/Chk2, p53, and Cdc25.

Q13. Explain the molecular pathology of Li-Fraumeni Syndrome and the dominance of dominant-negative TP53 mutations in human carcinogenesis.

Q14. Discuss the biochemical mechanisms of Mitotic Catastrophe and explain how anti-mitotic chemotherapy drugs trigger apoptotic cell death.

Q15. Explain the structural mechanics of Condensin I and II complexes in hierarchical chromosome condensation during prophase.

Q16. Describe the role of Polo-Like Kinase 1 (Plk1) as a master orchestrator of mitotic transitions.

Q17. Explain the molecular etiology and chromosomal consequences of Meiotic Non-Disjunction, contrasting maternal age-dependent cohesin decay.

Q18. Describe the structural organization and function of the Midbody during cytokinesis and explain how the NoCut checkpoint delays abscission.

Q19. Explain the mechanisms of Mitophagy mediated by the PINK1-Parkin pathway in cellular quality control and Parkinson's disease.

Q20. Discuss the molecular regulation of the Spindle Position Checkpoint (SPOC) and the Centrosome Duplication cycle.

Q21. Explain how the SCF (Skp1-Cullin-F-box) ubiquitin ligase complex regulates cell cycle transitions and contrast its substrate targeting with the APC/C.

Q22. Describe the biochemical structure and cellular functions of the Cohesin and Condensin complexes, highlighting the Smc-Kleisin architecture.

Q23. Explain the molecular regulation of the Anaphase-Promoting Complex switch from APC/C-Cdc20 to APC/C-Cdh1 during mitotic exit.

Q24. Discuss the biochemical mechanisms of Cellular Senescence, contrasting replicative senescence (telomere attrition) with oncogene-induced senescence (OIS).

Q25. Explain the role of the Kinetochore-Localized Ndc80 and Ska1 Complexes in coupling chromosome movement to microtubule depolymerization.

Q26. Describe the molecular mechanisms of the G2/M DNA Damage Checkpoint, detailing the roles of Chk1, Claspin, and 14-3-3 sigma.

Q27. Explain how the Anaphase-Promoting Complex (APC/C) recognizes its substrates via Destruction Boxes (D-Box) and KEN-Boxes.

Q28. Describe the molecular mechanisms of Chromosome Non-Disjunction leading to Aneuploidy and explain the resulting genomic instability in cancer.

Q29. Explain how the Aurora Kinase family (Aurora A, B, and C) coordinates centrosome maturation, spindle assembly, and cytokinesis.

Q30. Discuss the molecular composition of the Synaptonemal Complex and explain how the MutL-gamma resolvase complex executes crossover resolution.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[B]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[C]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[B]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[B]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[C]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[B]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[C]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

promoters.
3. Mitogenic Signaling and Initial Phosphorylation:
- Extracellular growth factors (EGF, PDGF) trigger the Ras-Raf-MEK-ERK signaling cascade, inducing transcriptional expression of D-type cyclins (Cyclin D1, D2, D3).
- Cyclin D binds CDK4 and CDK6, phosphorylating Rb at specific C-terminal serine residues (Ser-780, Ser-795).
- Initial phosphorylation causes Rb to dissociate from HDACs, relieving active repression, allowing low-level basal transcription of E2F target genes, notably Cyclin E and Cyclin A.
4. The Hyper-Phosphorylated Bistable Switch:
- Newly synthesized Cyclin E complexes with CDK2. Cyclin E-CDK2 phosphorylates Rb at multiple additional serine/threonine sites (> 14 sites total), fully hyper-phosphorylating Rb.
- Hyper-phosphorylated Rb undergoes an intramolecular domain clasp between its N-terminus and pocket domain, completely destroying its binding affinity for E2F.
- Free E2F rapidly drives massive transcription of S-phase genes: DNA Polymerase alpha, PCNA, MCM helicase, Thymidine Kinase, and crucially, Cyclin E itself!
- This creates two self-reinforcing, irreversible positive feedback loops: (1) E2F transactivates its own gene; (2) E2F drives Cyclin E synthesis, which activates CDK2, which hyper-phosphorylates more Rb to free more E2F.
- Once total Rb phosphorylation exceeds ~80%, the system flips into an autonomous, bistable 'ON' state that can no longer be reversed by mitogen withdrawal.

Solution to Question 3:

damage; p27 is regulated by TGF-beta signaling and contact inhibition, and is targeted for ubiquitin-mediated proteolysis by SCF-Skp2 upon mitogenic restimulation.

- Myt1 Kinase phosphorylates adjacent Threonine-14 (Thr-14).
- These negative charges sterically interfere with ATP phosphate orientation, holding MPF inactive despite activating Thr-161 phosphorylation by CAK.

2. Triggering Mitotic Entry:
As cells achieve critical cell mass and complete DNA replication, Plk1 (Polo-Like Kinase 1) is activated:
- Plk1 phosphorylates the dual-specificity phosphatase Cdc25C, partially activating it, and phosphorylates Wee1, marking it for degradation.
- Active Cdc25C dephosphorylates Thr-14 and Tyr-15 on CDK1, generating a small initial pool of active Cyclin B-CDK1.

3. Two Mutual Positive Feedback Loops (The Hysteresis Engine):
Once this initial active CDK1 appears, it drives two explosive, interlocking positive feedback loops:
- Positive Feedback Loop 1 (CDK1 -> Cdc25C): Active CDK1 directly phosphorylates Cdc25C on multiple N-terminal regulatory serines, increasing its phosphatase activity by > 20-fold! More active Cdc25C rapidly dephosphorylates more CDK1.
- Double-Negative (Positive) Feedback Loop 2 (CDK1 -| Wee1): Active CDK1 directly phosphorylates Wee1 kinase on its N-terminal regulatory domain, inactivating its kinase function and creating a phosphodegron recognized by SCF-beta-TrCP for rapid ubiquitin-proteasome destruction.

4. Greatwall Kinase and Phosphatase Suppression (The Ensa/PP2A Switch):
In interphase, the counteracting phosphatase PP2A-B55 continuously dephosphorylates CDK1 substrates, preventing mitotic entry:
- Active CDK1 phosphorylates and activates Greatwall Kinase (GWL / MASTL).
- GWL phosphorylates the small heat-stable proteins Ensa and ARPP19.
- Phosphorylated Ensa acts as an extraordinarily tight, stoichiometric inhibitor of PP2A-B55, completely shutting off substrate dephosphorylation.

5. Systems Consequence (Bistable Hysteresis):
This systems architecture converts a graded, continuous input (cyclin B accumulation) into an irreversible, all-or-none, bistable switch with zero intermediate states, driving the instantaneous, genome-wide execution of mitosis.

>> **Solution to Question 6:** Describe the structural organization and function of the Kinetochore, detailing the inner KMN network and the Ndc80 complex.

(SAC), presenting multiple MELT repeats that recruit Bub1 and Bub3 when phosphorylated by Mps1 kinase.

3. The Ndc80 Complex (The Primary Microtubule Coupler):

A 57 nm elongated, rigid heterotetramer consisting of two heterodimers joined in an antiparallel coiled-coil rod:

- Head Domain (Microtubule-Binding Face): Composed of the globular N-terminal Calponin Homology (CH) domains of Ndc80 (Hec1) and Nuf2.
- Tail Domain (Kinetochore-Facing): Composed of Spc24 and Spc25, forming a globular head that anchors into the Mis12 complex.
- The Basic N-terminal Tail: The extreme N-terminus of Ndc80 contains an intrinsically disordered, basic tail (~80 amino acids) carrying positive charges that bind electrostatically to the acidic, negatively charged C-terminal 'E-hooks' of alpha- and beta-tubulin protofilaments.
- Cooperativity: Multiple Ndc80 complexes oligomerize along a single microtubule plus-end, forming a flexible multivalent sleeve capable of maintaining mechanical attachment during both polymerization and curling depolymerization.

>>Solution to Question 7: Explain the Error Correction mechanism of Kinetochore-Microtubule Attachments mediated by Aurora B Kinase and Tension Sensing.

Constitutive phosphatases (PP1 and PP2A-B56) dephosphorylate the Ndc80 tail.
Dephosphorylated Ndc80 binds microtubules with maximum affinity, locking the attachment and satisfying the spindle assembly checkpoint.

Once the last kinetochore achieves amphitelic bipolar attachment, the Spindle Assembly Checkpoint is silenced:
- MCC production ceases, and free Cdc20 binds the catalytic core of the APC/C (specifically the Apc10 co-receptor and cullin-RING subunits Apc2/Apc11).
- This forms the active E3 ubiquitin ligase holoenzyme APC/C-Cdc20.
2. Targeted Polyubiquitination of Securin and Cyclin B:
- APC/C-Cdc20 recognizes conserved degradation motifs (D-Box: RXXLXXXXN; KEN-box: KENXXXN) on two critical targets:
- Securin (PTTG1): The stoichiometric inhibitory chaperone of Separase.
- Cyclin B1: The regulatory subunit of MPF (CDK1).
- In cooperation with E2 enzymes (UbcH10 and Ube2S), APC/C attaches Lys-48-linked polyubiquitin chains to Securin and Cyclin B, targeting them for rapid destruction by the 26S Proteasome.
3. Liberation and Activation of Separase:
- In metaphase, the endopeptidase Separase is completely inactive due to two independent inhibitory locks:
- Lock 1: Physical stoichiometric binding by Securin.
- Lock 2: Inhibitory phosphorylation on Ser-1126 by Cyclin B-CDK1, promoting complex formation with Cyclin B-CDK1.
- APC/C-mediated degradation of BOTH Securin and Cyclin B simultaneously dismantles both locks!
- Separase is liberated into its active, open protease conformation.
4. Cohesin Cleavage and Chromatid Disjunction:
- Centromeric sister chromatid cohesion is mediated by the ring-shaped Cohesin complex (Smc1, Smc3, Scc1/Rad21, Scc3).
- Active Separase performs site-specific endoproteolytic cleavage of the kleisin subunit Scc1 (Rad21) at two invariant consensus arginine residues (Arg-172 and Arg-450 in humans).
- Cleavage breaks the topological ring; the cohesin gate springs open, releasing the entrapped sister chromatid DNA strands.
- Under the pre-existing mechanical tension of kinetochore microtubules, the sister chromatids instantly disjoin and accelerate toward opposite spindle poles at ~1 micron/minute (Anaphase A).

>> **Solution to Question 10:** Describe the stages of Meiosis I and II, highlighting the molecular architecture of the Synaptonemal Complex and Chiasma resolution.

breaks down; spindle forms.
3. Architecture of the Synaptonemal Complex (SC):
A tripartite, ladder-like macromolecular protein scaffold:
- Two Lateral Elements (LEs): Formed along the axial cores of each homolog, composed of SYCP2 and SYCP3, anchoring cohesin complexes containing meiotic kleisin Rec8.
- Central Element (CE): Located midway across a 100 nm gap, assembled from SYCE1, SYCE2, SYCE3, and TEX12.
- Transverse Filaments (TFs): Formed by homodimers of SYCP1 oriented in an antiparallel zip-like fashion, with their C-termini anchored in the lateral elements and their N-termini interdigitating in the central element.
4. Meiosis I vs. Meiosis II Segregation Logic:
- In Anaphase I: Kinetochores of sister chromatids are co-oriented toward the SAME spindle pole (mono-orientation driven by the Meikin complex). Shugoshin-2 (Sgo2) recruits PP2A phosphatase to centromeric Rec8 cohesin, shielding it from separase cleavage.
- Separase cleaves ONLY arm cohesin, resolving chiasmata and allowing homologous chromosomes to separate to opposite poles, while sister chromatids remain linked at centromeres!
- In Anaphase II: Equational division. Kinetochores bi-orient; Sgo2 is absent; separase cleaves the remaining centromeric Rec8, allowing sister chromatid disjunction into four genetically unique haploid gametes.

>>Solution to Question 11 Explain the molecular mechanics of Cytokinesis in animal cells, detailing Centralspindlin, ECT2, RhoA, the Contractile Ring, and ESCR

helical spirals that narrow the midbody neck away from the cytoplasm.
- The microtubule-severing enzyme Spastin is recruited to clear microtubules out of the bridge.
- Further constriction and scission by ESCRT-III and the AAA+ ATPase Vps4 physically severs the membrane, achieving final Abscission and yielding two independent daughter cells.

>>Solution Question 12 the DNA Damage Checkpoint is a signaling cascade mediated by ATM/ATR, Chk1/Chk2, and TP53 mutations in human carcino

leukemias (the 'SBLA' cancer spectrum).

2. The Unique Mutation Profile of p53:

Unlike classic tumor suppressors (e.g., RB1, APC, BRCA1) which are inactivated by nonsense, frameshift, or deletion mutations leading to protein absence:

- Over 75% of p53 mutations in cancer are single amino acid Missense Mutations clustering within the sequence-specific DNA-Binding Domain (DBD, residues 102-292).
- Six 'Hotspot' residues account for > 30% of all mutations: Arg-175, Gly-245, Arg-248, Arg-249, Arg-273, Arg-282.
- Contact Mutants (e.g., R248, R273): Alter amino acids directly contacting the DNA phosphate backbone or major groove.
- Conformational / Structural Mutants (e.g., R175, G245): Disrupt local folding or coordinate zinc binding, destabilizing the beta-sandwich scaffold.

3. The Dominant-Negative Mechanism:

- Active p53 must assemble as a homotetramer (a dimer of dimers) to bind its consensus DNA response elements.
- In heterozygous cells carrying one wild-type and one missense mutant allele, statistical subunit mixing produces tetramers containing 0, 1, 2, 3, or 4 mutant monomers.
- Incorporation of even a single mutant subunit into the tetrameric complex distorts the spatial orientation of the DNA-binding surfaces, inactivating the entire tetramer!
- Out of 16 possible tetramer configurations, only 1/16 (6.25%) is composed of pure wild-type subunits. Thus, the mutant allele exerts a dominant-negative poisoning effect over the remaining wild-type allele even before Loss of Heterozygosity (LOH) occurs.

4. Gain-of-Function (GOF) Oncogenic Properties:

Mutant p53 proteins escape MDM2-mediated degradation and accumulate to massive levels. Many missense mutants physically interact with other transcription factors (p63, p73), inhibiting their pro-apoptotic functions, or bind novel genomic loci to upregulate pro-metastatic genes, conferring drug resistance and invasive properties.

>> **Solution to Question 14:** Discuss the biochemical mechanisms of Mitotic Catastrophe and explain how anti-mitotic chemotherapy drugs trigger apoptotic cell dea

4. The Two Competing Molecular Clocks (Bcl-2 vs. Cyclin B):
During prolonged mitotic arrest, two independent, competing biochemical degradation timers run concurrently:
- Death Clock (Apoptotic Priming): Active CDK1 phosphorylates anti-apoptotic proteins Mcl-1 and Bcl-xL, targeting Mcl-1 for ubiquitin-mediated degradation by the SCF-FBW7 complex. Concurrently, pro-apoptotic BH3-only proteins (Bim, Noxa) accumulate.
- Survival Clock (Mitotic Slippage): Despite SAC engagement, a slow, continuous leak of Cyclin B degradation occurs via residual APC/C activity ('mitotic slippage').
- Outcome: If Mcl-1 levels fall below the critical survival threshold before Cyclin B drops, Bax/Bak are activated, inducing MOMP and explosive caspase-dependent apoptosis directly from mitotic arrest!
- If mitotic slippage occurs first, the cell exits mitosis without division, becoming a tetraploid G1 cell that is subsequently cleared by p53-dependent senescence or apoptosis.

>>Solution to Question 16: Explain the structural mechanics of Condensin I and II complexes in hierarchical chromosome condensation during prophase.

[Problem Statement]: Describe the role of Polo-Like Kinase 1 (Plk1) as a master orchestrator of mitotic transitions.
[Detailed Step-by-Step Answer]:
Polo-Like Kinase 1 (Plk1) Orchestration Across Mitosis:
1. Structural Organization of Plk1:
Plk1 consists of an N-terminal Ser/Thr kinase domain and a unique C-terminal non-catalytic Polo-Box Domain (PBD) composed of two polo boxes:
- The PBD acts as an intramolecular inhibitory clamp that folds back over the kinase domain in basal state.
- Priming Phosphorylation: Upstream kinases (e.g., CDK1) phosphorylate substrates on Ser-[pSer/pThr]-Pro motifs. The PBD recognizes and binds this pre-phosphorylated docking motif, relieving auto-inhibition and targeting Plk1 precisely to specific subcellular structures (centrosomes, kinetochores, spindle midzone).
2. Five Master Mitotic Transitions Governed by Plk1:
- Centrosome Maturation and Separation (G2/M): Plk1 phosphorylates pericentriolar material (PCM) scaffolding proteins (pericentrin, NEDD1), recruiting gamma-tubulin ring complexes to expand microtubule-nucleating capacity. Concurrently, it activates Nek2 to sever the intercentriolar protein tether, driving centrosome separation.
- Mitotic Entry Trigger: Plk1 phosphorylates Cdc25C on Ser-198 to activate its phosphatase function and phosphorylates Wee1 to mark it for SCF-beta-TrCP degradation, igniting the explosive activation of MPF.
- Cohesin Removal via the Prophase Pathway: Phosphorylates the SA2 (Scc3) subunit of cohesin along chromosome arms, enabling Wapl to release cohesin rings from DNA without separase cleavage.
- Kinetochore-Microtubule Attachment: Phosphorylates Bub1, BubR1, and KNL1, recruiting checkpoint components and stabilizing amphitelic attachments.
- Cytokinesis Initiation: During anaphase, Plk1 localizes to the central spindle, phosphorylating the MgcRacGAP subunit of Centralspindlin, creating the recruitment docking site for the RhoGEF ECT2 to activate RhoA.

>> **Solution to Question 17:** Explain the molecular etiology and chromosomal consequences of Meiotic Non-Disjunction, contrasting maternal age-dependent cohes

factor Sgo2 are loaded during fetal S-phase and CANNOT be renewed or resynthesized during adulthood.

- Over decades of dictyate arrest, meiotic cohesin rings undergo gradual, spontaneous oxidative and proteolytic deterioration ('the Cohesin Aging Hypothesis').
- Progressive loss of arm cohesin causes premature chiasma terminalization and dissociation of homologous bivalents into unpaired univalents.
- Loss of centromeric cohesin allows sister kinetochores to drift apart and bi-orient prematurely in Meiosis I instead of Meiosis II, causing catastrophic random chromosome missegregation upon ovulatory maturation.

>>Solution Question 18: Describe the mechanism of Mitophagy mediated by the PINK1-Parkin pathway in cellular quality control and Parkinson's disease absci

2. The Voltage-Sensing Mechanism of PINK1:
PINK1 (PTEN-Induced Kinase 1) is a Ser/Thr kinase that acts as an electrochemical voltmeter of the inner mitochondrial membrane:
- In Healthy Mitochondria (High Membrane Potential Delta Psi): Newly synthesized PINK1 is continuously imported across both membranes via TOM and TIM23 complexes. The inner membrane protease PARL cleaves PINK1 within its transmembrane domain, releasing a 52 kDa truncated fragment that diffuses into the cytosol and is rapidly degraded by the N-end rule ubiquitin-proteasome pathway. Healthy mitochondria maintain undetectable basal PINK1!
- In Damaged/Depolarized Mitochondria (Delta Psi Collapsed): Loss of the proton motive force halts electrophoretic import through TIM23. PARL cannot cleave PINK1. Full-length, active 64 kDa PINK1 accumulates on the outer mitochondrial membrane (OMM) as an active homodimeric complex associated with TOM40.
3. Parkin Recruitment and Ubiquitin Phosphorylation:
- Accumulated PINK1 autophosphorylates and phosphorylates pre-existing Ubiquitin molecules attached to OMM proteins at a conserved serine residue: Ser-65 (forming Phospho-Ubiquitin).
- Phospho-Ubiquitin serves as an extraordinarily high-affinity receptor that recruits the cytosolic E3 ubiquitin ligase Parkin (PRKN) from the cytoplasm to the damaged mitochondrion.
- PINK1 directly phosphorylates the ubiquitin-like (UBL) domain of Parkin (at Ser-65), relieving Parkin's auto-inhibition and activating its E3 ligase activity.
4. Feed-Forward Ubiquitin Amplification:
Active Parkin decorates numerous outer membrane proteins (Mfn1, Mfn2, VDAC, Miro) with dense polyubiquitin chains. PINK1 phosphorylates these newly added ubiquitins, recruiting more Parkin, creating a rapid, self-amplifying feed-forward loop.
5. Autophagosomal Clearance:
Phospho-ubiquitin chains recruit autophagy adaptor proteins (OPTN, NDP52, p62) via their ubiquitin-binding domains. Adaptors bind LC3-II on expanding isolation membranes, engulfing the damaged mitochondrion into an autophagosome targeted for lysosomal degradation.
6. Clinical Connection:
Autosomal recessive loss-of-function mutations in PINK1 (PARK6) or PRKN (PARK2) cause early-onset familial Parkinson's disease due to progressive accumulation of damaged mitochondria and ROS-induced death of dopaminergic neurons in the substantia nigra.

Solution to Question 20:

>> Discuss the molecular regulation of the Spindle Position Checkpoint (SPOC) and the Centrosome Duplication cycle.

In asymmetric cell divisions (e.g., budding yeast and mammalian stem cells), the mitotic spindle must align precisely along the polarity axis to ensure unequal partitioning of developmental determinants:

- The Kinase Signaling Cascade (Bub2-Bfa1): The GAP complex Bub2-Bfa1 localizes to spindle poles, maintaining the small GTPase Tem1 in an inactive GDP-bound state, preventing mitotic exit network (MEN) activation as long as the spindle is misaligned.
- Alignment Trigger: When the spindle properly aligns and the daughter spindle pole migrates across the bud neck into the daughter cell, Tem1 encounters its activator Lte1 (localized exclusively in the daughter cortex).
- Active Tem1-GTP triggers the protein kinase Cdc15 and Dbf2, driving release of Cdc14 phosphatase from the nucleolus to dephosphorylate CDK1 substrates, enabling exit from mitosis strictly when spatial orientation is correct.

---

1. The Conserved SMC Architectures:
Both Cohesin and Condensin are evolutionary descendants of prokaryotic SMC homodimers, sharing a conserved ring-like structural topology:
- SMC Heterodimers: Formed by two distinct SMC polypeptides (Smc1-Smc3 in Cohesin; Smc2-Smc4 in Condensin).
- Each SMC monomer folds back on itself: its central amino acid sequence forms an antiparallel coiled-coil rod (~50 nm long), bringing the N- and C-termini together to form an ATP-binding cassette (ABC-type) catalytic 'Head' at one end, and a flexible dimerization 'Hinge' at the other.
- Two SMC monomers heterodimerize at their hinges, while their heads are bridged by a non-SMC protein belonging to the Kleisin family, enclosing a large topological lumen capable of encircling DNA.
2. Cohesin Complex Specialization:
- Subunit Composition: Smc1, Smc3, Scc1 (Rad21 / Mcd1 kleisin), and Scc3 (STAG1/2).
- Topological Action: Cohesin acts as an intermolecular topological clamp that entraps TWO double-stranded DNA molecules (sister chromatids) inside its 35 nm lumen, holding them together from S-phase until anaphase disjunction.
- Loading and Unloading: Loaded onto DNA in telophase/G1 by the Scc2-Scc4 (NIPBL-MAU2) loading complex; dynamically removed in prophase by Wapl-Pds5, and definitively cleaved at anaphase onset by Separase.
3. Condensin Complex Specialization:
- Subunit Composition: Smc2, Smc4, CAP-H (kleisin in Condensin I) or CAP-H2 (kleisin in Condensin II), and two HEAT-repeat subunits.
- Topological Action: Condensin acts as an INTRA-molecular loop-extruding motor. Instead of embracing two sister chromatids, a single condensin ring reels in a single double-stranded DNA fiber, generating expansive loops that drive 10,000-fold axial compaction of the chromosome into a rigid mitotic rod.

>> Solution to Question 23 Explain the molecular regulation of the Anaphase-Promoting Complex switch from APC/C-Cdc20 to APC/C-Cdh1 during mitotic exit.

(and PP2A-B55) to dephosphorylate Cdh1.
- Step 4: Dephosphorylated Cdh1 can now bind the APC/C with high affinity via its C-terminal C-box and isoleucine-arginine (IR) tail, assembling active APC/C-Cdh1.
4. Autonomous Inactivation of Cdc20 by Cdh1:
Once formed, APC/C-Cdh1 recognizes a specific KEN-box on Cdc20 itself, targeting Cdc20 for polyubiquitination and destruction!
Thus, Cdh1 irreversibly eliminates its predecessor, cementing the transition and maintaining low cyclin levels throughout G1 until Cyclin E/A-CDK2 accumulates at the R-point to re-phosphorylate and shut off Cdh1.

---

translocation does NOT require ATP-fueled motor walking; instead, it directly harnesses the mechanical strain energy released by depolymerizing microtubule protofilaments!

2. The Protofilament Curling Power Stroke:

- The GDP-tubulin core of a microtubule is held in a strained, straight lattice by the GTP cap.
- Upon cap loss, protofilaments spring outward, peeling back into curved structures with a radius of curvature of ~20 nm.
- This conformational curl generates substantial mechanical force (~30-65 pN per protofilament), pushing against kinetochore couplers.

3. The Ndc80 Complex as a Sliding Sleeve:

- Multiple Ndc80 heterotetramers form a multivalent, oligomeric cluster surrounding the microtubule end.
- Ndc80 CH-domains bind weakly and reversibly to adjacent tubulin monomers.
- As outward-curling protofilaments peel back, they exert steric force against the Ndc80 heads, pushing the entire complex backward toward the minus-end without detachment (biased diffusion / power stroke coupling).

4. The Ska1 Complex (The Human Equivalent of the Dam1 Ring):

- The human Ska complex (Ska1 - Ska2 - Ska3) forms a V-shaped homodimer that binds directly to Ndc80 and possesses an unconventional curved winged-helix domain that tracks along the curved surface of peeling protofilaments.
- Ska1 dramatically increases the grip of Ndc80 on depolymerizing ends, reducing the rate of accidental detachment by 10-fold and allowing kinetochores to pull against heavy opposing viscous loads (up to 700 pN per chromosome).

>>**Solution to Question 26:** Describe the molecular mechanisms of the G2/M DNA Damage Checkpoint, detailing the roles of Chk1, Claspin, and 14-3-3 sigma,

- Active MPF (Cyclin B-CDK1) cannot assemble, locking the cell in stable, prolonged G2 arrest until repair enzymes complete lesion excision.

---

nuclear envelope reassembly, forming a separate, aberrant Micronucleus.
- Micronuclear envelopes have defective nuclear pore complexes and fragile lamina, frequently undergoing catastrophic spontaneous rupture during interphase.
- Cytosolic nucleases (TREX1) and premature replication in ruptured micronuclei shatter the trapped chromosome into hundreds of fragments.
- Non-Homologous End Joining (NHEJ) randomly stitches the fragments back together in a scrambled orientation, a catastrophic single-event chromosomal shattering phenomenon termed Chromothripsis, driving massive oncogenic amplification and loss of tumor suppressors.

- Transverse Filaments (TFs): Transverse protein SYCP1 homodimers extend across the 100 nm central space. Their C-termini anchor into the lateral elements, while their coiled-coil N-termini meet in the center.
- Central Element (CE): A longitudinal lattice of SYCE1, SYCE2, SYCE3, and TEX12 that polymerizes to stabilize the zipped transverse filaments.

2. Double-Strand Break Repair Pathway (The Canonical Crossover Pathway):
- Programmed DSBs introduced by Spo11 are resected to generate 3' single-stranded DNA tails.
- Meiosis-specific recombinase Dmc1 and canonical Rad51 catalyze strand invasion into the non-sister homologous chromatid, forming a displacement loop (D-loop).
- Second-end capture and DNA synthesis form a Double Holliday Junction (dHJ) intermediate.

3. Biased Crossover Resolution by MutL-gamma (Mlh1-Mlh3 Resolvase):
In mitotic DNA repair, Holliday junctions are non-crossover resolved to prevent genomic rearrangements. In contrast, meiosis strictly requires at least one Crossover per chromosome (the 'Obligate Crossover') to establish physical chiasmata for accurate segregation:
- The heterodimeric MutS-gamma complex (Msh4-Msh5) binds and stabilizes the double Holliday junction.
- Msh4-Msh5 recruits the MutL-gamma endonuclease heterodimer, composed of Mlh1 and Mlh3.
- Mlh3 contains a conserved metal-dependent DQHA(X)2E(X)4E nuclease motif.
- Guided by the pro-crossover factor Exo1 and polymerizing PCNA clamps, MutL-gamma introduces asymmetric, coordinated single-strand nicks into two non-crossing strands of the dHJ, resolving the topological junctions exclusively into Crossover recombinant products with flanking chiasmata.
